

## OTHER GRADE PAPER

# Frameworks to identify research gaps, frame research needs, and derive research priorities from evidence syntheses: a scoping review

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## Abstract

**Objectives:** To identify and describe frameworks for systematically identifying research gaps from evidence syntheses, framing research needs, and establishing priorities, with emphasis on their use of GRADE (Grading of Recommendations Assessment, Development and Evaluation) certainty of evidence (CoE) domains and evidence-to-decision (EtD) criteria.

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**Study Design and Setting:** We conducted a scoping review following JBI guidance, searching MEDLINE, PsycINFO, and Web of Science on July 14, 2025, for publications from 2011 onward. Eligible documents described approaches for identifying research gaps or formulating research needs from systematic reviews or other evidence syntheses, not limited to GRADE. Data were extracted on framework structure, methods for gap identification, use of GRADE CoE to explore nature of the gaps, GRADE EtD to frame needs, and strategies for prioritization.

**Results:** Fourteen frameworks from 17 publications were identified. Five frameworks used the Population, Interventions, Comparisons and Outcomes (PICO) approach to characterize research gaps. Six frameworks provided explicit methods to investigate the nature of the gaps, typically through GRADE CoE assessment. None fully integrated the GRADE EtD framework, although some considered selected EtD criteria when framing research needs. Four included methods for prioritizing needs.

**Conclusion:** Frameworks to identify research gaps and formulate needs from evidence syntheses are heterogeneous and often lack explicit methods for investigating the nature of the gaps or prioritizing research. This review will inform guidance on how to use GRADE to systematically and transparently inform future research. © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

**Keywords:** Research recommendations; Research gaps; GRADE; Evidence-based medicine; Systematic reviews; Evidence synthesis; Health technology assessment

## 1. Introduction

Evidence-based medicine rests on two essential functions: synthesizing what is known and identifying what remains unknown [1,2]. While systematic reviews and practice guidelines excel at summarizing the best available evidence [3], the way authors currently identify and report research gaps is often inconsistent and insufficient [4–7]. Sections on future research in reviews and practice guidelines are frequently brief, generic, and lack the specificity needed to guide meaningful follow-up studies [6,7] (manuscript in preparation).

Several priority-setting frameworks to address research gaps and needs exist such as the James Lind Alliance (JLA) Priority Setting Partnerships [8] and the Child Health and Nutrition Research Initiative (CHNRI) [9]. However, they operate independently of the evidence synthesis process. This disconnect limits the ability to fully leverage systematic reviews and guidelines to inform future research. The absence of a structured link between synthesis outputs and research priorities has important implications for the evidence and decision-making ecosystem. For researchers, it represents missed opportunities to design studies that directly address the most pressing gaps while avoiding redundancy and resource waste. For funders, it creates uncertainty about which unanswered questions are most critical, increasing the risk of inefficient resource allocation or research duplication. For patients and policymakers, it delays the availability of evidence needed to improve health outcomes and guide informed decision-making.

The GRADE (Grading of Recommendations Assessment, Development and Evaluation) approach promotes the generation of transparent and structured outputs, such as Summary of Findings (SoF) tables [10] and evidence-to-decision (EtD) frameworks [11], which provide detailed assessments of both the certainty of evidence (CoE) and judgements regarding broader considerations that influence

health recommendations in guidelines and decisions, such as in health technology assessment (HTA) [12]. These outputs could be used to explicitly identify current gaps and prioritize future research [13]. Although we previously described how the GRADE CoE approach may be used to describe implications for research [13], there is no formal GRADE guidance.

To inform the development of GRADE guidance for generating explicit, evidence-based recommendations for future research based on the CoE assessment and EtD frameworks, we conducted a scoping review of structured frameworks designed and used to identify research gaps directly from evidence syntheses. Based on the findings and iterative discussions within the authorship team, we also propose a preliminary table to serve as an interim reference for future methodological development.

## 2. Methods

The protocol for the review has been published on Open Science Framework (OSF) [14]. The following manuscript describes the results of aim 3 of the protocol. The results for aims one, two, and four will be described elsewhere. The scoping review was designed following the JBI 2020 guide to scoping reviews [15]. Reporting adheres to the PRISMA Extension for Scoping Reviews (PRISMA-ScR) (Supplementary Appendix I) [16]. Recognizing the variability in the terminology used to define research gaps and needs [4], we developed definitions iteratively through multiple rounds of feedback to facilitate interpretation and conceptualize the relationship between the terms (Box 1 and Fig 1). During analysis, it became evident that little guidance exists to support groups in systematically identifying research gaps. To address this, we extended our protocol to develop a preliminary table that acts as a tool to

**What is new?****Key findings**

- This scoping review identified 14 frameworks to systematically identify research gaps, frame research needs, and establish priorities from evidence syntheses.

**What this adds to what is known?**

- The absence of a structured link between synthesis outputs and research priorities has important implications for the evidence and decision-making ecosystem.
- There is heterogeneity among available framework to identify research gaps and they poorly integrated explicit methods for investigating the nature of the gaps and prioritizing research needs.

**What is the implication and what should change now?**

- Future systematic reviews and guidelines should adopt explicit, structured methods to describe the rationale, nature, and priority of research gaps.
- We propose a preliminary table that review authors could use to aid them in the process.

identify research gaps, formulate research needs, and establish research priorities.

**2.1. Eligibility criteria**

In brief, we included studies, reports, or guidance documents that described or applied structured frameworks, tools, or methodological approaches to identify research gaps, define research needs, or derive research priorities directly from evidence syntheses without restriction by setting or health topic. These terms were used broadly to ensure an inclusive approach while recognizing variability in how such methods are described in the literature. Documents did not need to explicitly self-identify as frameworks; rather, any structured approach that could be operationalized to identify gaps, needs, or priorities was eligible. We excluded documents that discussed research prioritization outside of the evidence synthesis process such as the JLA priority setting partnership, which have already been comprehensively reviewed [4,5] and are currently being investigated elsewhere [17].

**2.2. Literature search**

We developed a search strategy in collaboration with an author with experience in bibliographic searches (SGL) to identify methods used to determine research gaps and

**Box 1 Conceptualization of research terms according to the authors.**

**Evidence or research gap:** A limitation, deficiency, or unexplored avenue in the body of available studies or synthesized evidence that restricts certainty in estimates across one or more GRADE domains. Such gaps may arise from weaknesses that reduce certainty (eg, risk of bias, inconsistency, indirectness, imprecision, or publication bias) or from insufficient exploration of factors that could increase certainty (eg, potential dose–response gradients, signals of large effects, or plausible residual confounding). These gaps may also arise from entirely missing evidence or from existing studies that inadequately address critical aspects of the research question (eg, population, intervention, comparator, outcomes, or contextual factors). Evidence gaps restrict the robustness of conclusions and often inform research prioritization [22,23].

**Research need:** A research need can be defined as an evidence gap judged to be directly relevant for decision-making, where addressing the uncertainty would meaningfully strengthen the evidence base for clinical or policy choices. Such a need encompasses two interrelated elements: first, a value judgment that the existing uncertainty is consequential enough to warrant reduction in order to improve the quality of decisions; and second, a specification of the research action required, delineating what type of study should be undertaken and how it should be conducted to appropriately resolve the gap—for example, the design, population, follow-up duration, outcomes, or measurement tools. Research needs can be systematically mapped to the GRADE EtD domains, including health effects, cost-effectiveness, equity, feasibility, and acceptability. Not all evidence gaps rise to the level of research needs; only those that carry clear implications for decision-making and are accompanied by actionable research specifications should be classified as such.

**Research priority:** A research need considered sufficiently important and urgent to warrant focused investigation and resource allocation. A research priority represents an explicit judgment about which need should be addressed first, recognizing that while all priorities are research needs, not all needs are elevated to priorities.

formulate research needs (Supplementary Appendix II). Searches were conducted on July 14th, 2025, in MEDLINE via Ovid, PsycINFO via EBSCOhost, and Web of Science, limited to publications from 2011 onward to capture recent methodological developments following the comprehensive



**Figure 1.** Conceptual relationship between research gaps, research needs, and research priorities.

systematic search by one member of our author team (Robinson et al [18,19]). To ensure we did not miss published frameworks prior to 2011, we conducted forward and backward citation searches of relevant reviews [4,5] using citationchaser [20]. Search terms combined controlled vocabulary (e.g., MeSH) and free-text keywords related to research priorities, research gaps, and priority setting. Duplicates were removed using EndNote (version 20.4.1) as described by Bramer et al [21]. In addition, members of the GRADE working group and invited members of the Evidence-Based Research Network (EBRN) were prompted to provide relevant resources and frameworks, including unpublished or gray literature sources.

### 2.3. Study selection and data extraction

Screening was done on Covidence software (Veritas Health Innovation Ltd, AU). Two of the following set of reviewers (O.D., M.E., M.B., P.J., K.M.R., M.L.) independently screened all titles and abstracts, and full-text documents to determine eligibility. Conflicts at any stage were resolved through discussion or via a third reviewer (M.L.). Before data extraction was undertaken, the form was pilot tested by two reviewers (O.D. and M.E.) on a sample of three frameworks and refined accordingly. The remaining data extraction was done by a single reviewer (O.D.) using a piloted standardized form and validated by a second reviewer (M.E. or M.R.). From each included study, we extracted: i) bibliographic details, ii) key descriptive information on the framework's objectives, iii) overall structure and steps of the framework, iv) approach to gap identification (ie, what to look at), v) classification for the nature of research gap (ie, cutoff or rationale for what

makes it qualify as a research gap), vi) how gaps translate into research needs, and vii) how they should be prioritized.

We recorded whether and how the framework explicitly applied GRADE CoE domains (limitations in study design or execution, inconsistency, indirectness, imprecision, dissemination bias) to justify the research gap. We also noted whether it used EtD criteria (eg, desirable and undesirable health effects, values, resource use, equity, acceptability, feasibility) to translate the gaps into needs.

We also documented any prioritization methods to address research needs included in the processes of developing research recommendations (terms explained in Box 1).

Risk of bias was not assessed following standard method of scoping reviews [16].

### 2.4. Data analysis and charting process

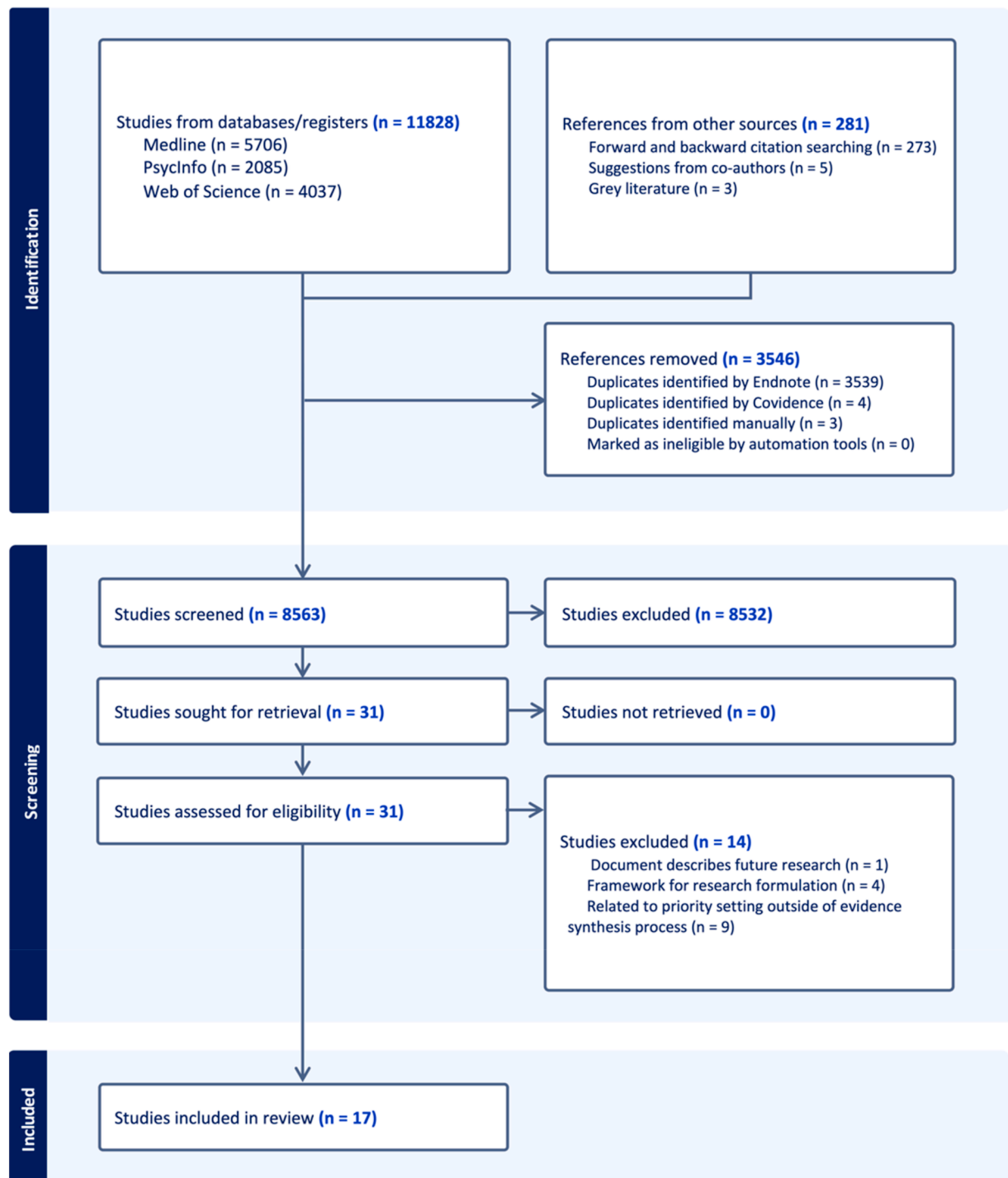
We summarized the characteristics of the included frameworks using frequencies and percentages, and described the features of the frameworks narratively and in tables. We charted extracted information according to four sequential domains: 1) approaches for assessing evidence and identifying research gaps, 2) reported the nature of gaps (eg, absence of evidence, low CoE), 3) translation of gaps into research needs, and 4) methods for prioritizing research needs. Figure was generated using ggplot2 in R [24].

### 2.5. Preliminary tool development

The development of the preliminary table was grounded in the findings of this scoping review. Drawing on elements from the included frameworks, we organized key features into a draft structure that logically reflected the steps of identifying research gaps, formulating research needs, and establishing priorities. Where appropriate, approaches were combined to improve utility and ensure alignment with current methodological guidance and the GRADE approach. The draft table was then reviewed and refined through iterative discussions among the authorship team, who contributed methodological expertise and experience in guideline development. Decisions regarding the inclusion, exclusion, and organization of content were reached by consensus, with the aim of producing a pragmatic and useable interim tool. The resulting table is presented as a preliminary resource to be further developed and validated through formal GRADE guidance processes [25].

## 3. Results

8563 records were screened at title and abstract stage from electronic search, forward and backward citation of relevant reviews, and gray literature resources suggested by team members. Thirty-one were screened at full text.



**Figure 2.** PRISMA flow chart.

Seventeen documents [6,7,13,19,26–38] corresponding to 14 independent frameworks [6,7,13,19,26–31,35–38] were included (Fig 2). No additional frameworks published in the gray literature were identified by GRADE working group or invited members.

### 3.1. Description of the included frameworks

Fourteen frameworks were identified, including one framework by National Institute for Health and Care Excellence (NICE) in the UK [28], another produced on behalf of the EBRN [36], a third one for the purpose of World Health

**Table 1.** Description of frameworks to inform research priorities from evidence syntheses

| Study ID                 | Framework structure                                                                                                                                                                                                                                                                                | Approach to identifying gaps                                                                                                                                                                                                                                                                                                         | Classifying the nature of the research gap                         | Integration of GRADE certainty of evidence domains to classify the nature of the research | GRADE EtD criteria to frame research needs                                                                                                                                                                                                                     | Methods to prioritize research gaps                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Annemans 2020 (TRUST4RD) | <ol style="list-style-type: none"> <li>1. Define evidence gaps according to area of uncertainty</li> <li>2. Select optimal data source and design to generate reliable evidence</li> <li>3. Conduct structured stakeholder engagement</li> </ol>                                                   | Categorize uncertainty into the following domains: <ul style="list-style-type: none"> <li>- Population size/ characteristics</li> <li>- Natural history/current management of the disease</li> <li>- The new treatment (effect size, safety, durability, subgroups, etc.)</li> <li>- Health care ecosystem and its actors</li> </ul> | Relies on identification approach to explain the nature of the gap | No integration of GRADE CoE domains                                                       | No explicit consideration of GRADE EtD framework: <ul style="list-style-type: none"> <li>- Desirable and undesirable effects</li> <li>- Importance of the problem</li> <li>- Values and preferences</li> <li>- Acceptability</li> <li>- Feasibility</li> </ul> | <ol style="list-style-type: none"> <li>1. Apply a multistakeholder deliberative process to score impact of each uncertainty.</li> <li>2. Combine quantitative impact estimation (e.g., modeling “what-if” scenarios for survival, QALYs, cost-offsets) with qualitative valuation from patient and clinician perspectives.</li> <li>3. Derive impact score scores to focus evidence generation on high-impact gaps and avoid low-value data collection</li> </ol> |
| Clarke 2007              | <ol style="list-style-type: none"> <li>1. Decide if a recommendation for research is needed</li> <li>2. Describe the research recommendation by specific population, intervention, outcome</li> <li>3. Suggest a new, expanded or updated systematic review</li> </ol>                             | Unclear how to decide if a recommendation for research is needed                                                                                                                                                                                                                                                                     | Not included in the approach                                       | Study predates GRADE CoE framework                                                        | Study predates GRADE EtD framework                                                                                                                                                                                                                             | No                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| De Vet 2001              | <ol style="list-style-type: none"> <li>1. Identify the number of randomized controlled trials (RCTs) in the specified computerized database</li> <li>2. If &gt; 5 RCTs are identified, recommend conducting a systematic review; if ≤ 5 RCTs are identified, proceed to review findings</li> </ol> | Research gap is determined by quantity of trials                                                                                                                                                                                                                                                                                     | Inconsistency of findings across trials                            | Study predates GRADE CoE framework (Inconsistency assessed)                               | Study predates GRADE EtD framework                                                                                                                                                                                                                             | None                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

(Continued)

Table 1. Continued

| Study ID     | Framework structure                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Approach to identifying gaps                                     | Classifying the nature of the research gap | Integration of GRADE certainty of evidence domains to classify the nature of the research | GRADE EtD criteria to frame research needs | Methods to prioritize research gaps |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------|
|              | 3. For each eligible trial, record study design (parallel or crossover), sample size, randomization status, blinding, interventions compared, and conclusions<br>4. If conclusions are inconsistent, recommend conducting a new RCT<br>5. If results are consistent, examine the evidence in detail to determine the level of certainty regarding effectiveness or ineffectiveness<br>6. When convincing evidence is found, integrate findings into clinical guidelines |                                                                  |                                            |                                                                                           |                                            |                                     |
| Johnson 2003 | Determine whether the primary clinical question was adequately answered by the available evidence by categorizing into one of the three categories:<br>(A) Meta-analysis provides conclusive evidence for benefit or harm<br>(B) Evidence is inconclusive, and review authors explicitly recommend further research<br>(C) Evidence is inconclusive, and no call for further research is needed                                                                         | Decide if evidence is inclusive but unclear how it is determined | Not included in the approach               | Study predates GRADE CoE framework                                                        | Study predates GRADE EtD framework         | None                                |

(Continued)

Table 1. Continued

| Study ID                | Framework structure                                                                                                                                                                                                                                                                                                                                                | Approach to identifying gaps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Classifying the nature of the research gap                                                                                                                                                                   | Integration of GRADE certainty of evidence domains to classify the nature of the research                                              | GRADE EtD criteria to frame research needs                                                                                                                  | Methods to prioritize research gaps                                                                                                                                                                                                                                                                                           |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lund 2021, EBR approach | <ol style="list-style-type: none"> <li>1. Decide if a definitive conclusion can be drawn</li> <li>2. Assess if there is confidence in this conclusion, or in other words, is the evidence of sufficiently high certainty</li> <li>3. formulate implications for clinical practice and suggestions for future research according to areas of uncertainty</li> </ol> | <ul style="list-style-type: none"> <li>- Absence of high certainty in the evidence</li> <li>- inability to draw conclusions from the evidence base</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                            | Not included in the approach                                                                                                                                                                                 | No integration of GRADE CoE domains                                                                                                    | No consideration of GRADE EtD criteria                                                                                                                      | None                                                                                                                                                                                                                                                                                                                          |
| NICE 2015               | <ol style="list-style-type: none"> <li>1. Identify, summarize the uncertainties and prioritize them</li> <li>2. Develop research recommendations using appropriate technique (e.g., PICO)</li> <li>3. Consult on research recommendations</li> <li>4. Review research recommendations as part of review cycle</li> </ol>                                           | <p>Identify uncertainties by the following categories:</p> <ul style="list-style-type: none"> <li>-Absence of evidence because</li> <li>-Research not done</li> <li>-Research done but not published</li> <li>-Research done and published but not identified</li> <li>-Evidence is available but               <ul style="list-style-type: none"> <li>- Selective reporting</li> </ul> </li> <li>- Methodological limitations</li> <li>- Inconsistency</li> <li>- Indirectness</li> <li>- Contextual challenges</li> <li>- Research outdated</li> </ul> | Relies on identification approach to explain the nature of the gap                                                                                                                                           | No explicit integration of GRADE CoE domains: risk of bias; imprecision; indirectness; inconsistency; dissemination (publication) bias | <p>No explicit consideration of GRADE EtD framework</p> <p>Considers:</p> <ul style="list-style-type: none"> <li>- Equity</li> <li>- Feasibility</li> </ul> | <ul style="list-style-type: none"> <li>- Interest-holders engagement to prioritize gaps including clinicians, researchers, patients and carers</li> <li>- Use of modeling exercises (e.g., economic modeling) to inform priorities</li> <li>- Additional analysis as needed (e.g., 'value-of-information' methods)</li> </ul> |
| Robinson 2011           | <ol style="list-style-type: none"> <li>1. Characterize of gaps using PICO framework</li> <li>2. Classify the nature(s) of the research gap using GRADE CoE domains</li> </ol>                                                                                                                                                                                      | PICO framework to identify gaps                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <ul style="list-style-type: none"> <li>- Insufficient or imprecise information</li> <li>- Biased information</li> <li>- Inconsistency or unknown consistency</li> <li>- Not the right information</li> </ul> | Explicit integration of GRADE CoE domains: risk of bias; imprecision; indirectness; inconsistency; dissemination (publication) bias    | Study predates GRADE EtD framework                                                                                                                          | None                                                                                                                                                                                                                                                                                                                          |

(Continued)

Table 1. Continued

| Study ID                                         | Framework structure                                                                                                                                                                                                                                                                                                                                                                                                                              | Approach to identifying gaps                                                                                                                                                                                                                                                                                     | Classifying the nature of the research gap                                                                                                                            | Integration of GRADE certainty of evidence domains to classify the nature of the research          | GRADE EtD criteria to frame research needs                                                                                                                         | Methods to prioritize research gaps                                                                                                                                                                                         |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Robinson 2011a                                   | 1. Identify of research gaps<br>2. Classify research gaps                                                                                                                                                                                                                                                                                                                                                                                        | Identification of research gap by:<br>- Type of management issue (comparative, long-term effects, implementation, clinical assessment)<br>- Specificity of target population (e.g., general population, clinical subgroups)<br>- Age of the target population as key factor<br>- PICO framework to identify gaps | - Insufficient or absence of evidence (to inform recommendations)<br>- Areas or topics needing further research (as identified by the guideline development exercise) | No integration of GRADE CoE domains                                                                | Study predates GRADE EtD framework<br>Considers:<br>- Desirable and undesirable effects<br>- Diagnostic test accuracy<br>- Values and preferences<br>- Feasibility | None                                                                                                                                                                                                                        |
| Saldanha 2013                                    | 1. Identify and abstract of research gaps<br>2. Collect feedback from authors of the review<br>3. Translate of research gaps into researchable questions<br>4. Collect feedback from local stakeholders (via online and in-person meetings)<br>5. Collect feedback from external stakeholders (via Delphi)<br>6. Prioritize of outcomes<br>7. Refine of research questions and development of conceptual model to display the research questions | PICO framework to identify gaps                                                                                                                                                                                                                                                                                  | Not included in the approach                                                                                                                                          | No integration of GRADE CoE domains                                                                | Study predates GRADE EtD framework<br>Considers:<br>- Desirable and undesirable effects<br>- Values and preferences<br>- Feasibility                               | 1. Feedback from local stakeholders<br>2. Feedback from external stakeholders<br>3. Prioritization of outcomes<br>4. Refinement of research questions and development of conceptual model to display the research questions |
| Schünemann 2019, Cochrane GRADEing Methods Group | 1. Define research using the PICO framework<br>2. Link to GRADE CoE domains<br>3. Specify the nature of further research needed                                                                                                                                                                                                                                                                                                                  | Use PICO framework to describe what's missing or could be improved                                                                                                                                                                                                                                               | Apply GRADE CoE assessments to highlight outcomes with low certainty and thus potential gaps                                                                          | Explicit integration of GRADE CoE domains: risk of bias; imprecision; indirectness; inconsistency; | No explicit consideration of GRADE EtD framework                                                                                                                   | None                                                                                                                                                                                                                        |

(Continued)

Table 1. Continued

| Study ID       | Framework structure                                                                                                                                                                                                                                                                                                     | Approach to identifying gaps                                                                                                                                                                                                                                                                 | Classifying the nature of the research gap                                                                                                                                                                                                | Integration of GRADE certainty of evidence domains to classify the nature of the research                                           | GRADE EtD criteria to frame research needs                                                  | Methods to prioritize research gaps                                                |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|                | 4. Develop outcome-specific research recommendations                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                           | dissemination (publication) bias                                                                                                    |                                                                                             |                                                                                    |
| Shepard 2007   | 1. Develop a care pathway to be used as a framework to map the literature<br>2. Conduct the evidence synthesis and map the literature to the pathway<br>3. Formulate the gaps as research or policy questions<br>4. Conduct prioritization exercise (Delphi and stakeholder engagement) to identify research priorities | Use of a care pathway as a framework; validated is preferred, otherwise devise and validate                                                                                                                                                                                                  | - Areas of the care pathway that lack appropriate guidelines/guidance or evaluation research<br>- Gaps noted within existing guidelines/guidance or evaluation research<br>- Poor-quality and out of date studies and guidelines/guidance | Study predates GRADE CoE framework (Risk of bias assessed)                                                                          | Study predates GRADE EtD framework                                                          | - Virtual workshop via email and Delphi                                            |
| Siegfried 2020 | 1. Assess of the GRADE strength and CoE rating of recommendations<br>2. Include of research questions identified from a global stakeholder survey<br>3. Select the GER standards and principles most relevant to the question through discussion and consensus                                                          | Identify research gaps based on the following certainty rules:<br>- Conditional or very low/ low certainty<br>- Moderate or high certainty missing GER principles<br>- No recommendation due to absence of evidence<br>- Strong recommendations from evidence only in high-resource settings | Relies on identification approach to explain the nature of the gap                                                                                                                                                                        | Explicit integration of GRADE CoE domains: risk of bias; imprecision; indirectness; inconsistency; dissemination (publication) bias | No explicit consideration of GRADE EtD framework: Equity Values and preferences Feasibility | Iterative discussion and ultimately consensus between guideline panel participants |
| Siemens 2023   | 1. Characterize the gaps using PICO framework and study design<br>2. Classify the nature of the research gap using GRADE and other approaches (e.g., research reporting)                                                                                                                                                | 1. Characterization of gaps using PICO framework and study design                                                                                                                                                                                                                            | methodological concepts; underlying GRADE domains to describe shortcomings of the body of evidence                                                                                                                                        | Explicit integration of GRADE CoE domains: risk of bias; imprecision; indirectness; inconsistency; dissemination (publication) bias | No consideration of GRADE EtD criteria                                                      | None                                                                               |

(Continued)

Table 1. Continued

| Study ID     | Framework structure                                                                                                                                                                                                                                                                                                     | Approach to identifying gaps                                      | Classifying the nature of the research gap                                                         | Integration of GRADE certainty of evidence domains to classify the nature of the research                                           | GRADE EtD criteria to frame research needs | Methods to prioritize research gaps |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|-------------------------------------|
| Siemens 2024 | <ol style="list-style-type: none"> <li>1. Characterize of gaps using PICO framework and study design</li> <li>2. Classify the nature of each research gap using GRADE and other approaches (e.g., research reporting)</li> <li>3. Decide if future research is no longer needed to improve the evidence-base</li> </ol> | 1. Characterization of gaps using PICO framework and study design | methodological concepts; underlying GRADE domains to describe shortcomings of the body of evidence | Explicit integration of GRADE CoE domains: risk of bias; imprecision; indirectness; inconsistency; dissemination (publication) bias | No consideration of GRADE EtD criteria     | None                                |

CoE, certainty of evidence; EtD, evidence to decision; GER, gender, equity and human rights; GRADE, Grading of Recommendations Assessment, Development and Evaluation; PICO, Population, Intervention, Comparator, Outcome; QALY, quality-adjusted life year.

Organization (WHO) recommendations [29], and three frameworks [19,37] developed under the auspices of the Evidence-based Practice Center Program of the Agency for Healthcare Research and Quality (AHRQ) (Table 1). The Cochrane Handbook for Systemic Reviews also included a framework assessing implications for practice from reviews on behalf of the Cochrane GRADEing methods group [13]. International collaborative groups authored the remaining frameworks.

Approaches ranged from including a single step [30] to involving seven [37]. Several frameworks were developed and piloted within specific clinical or policy contexts, including for rare disease treatments [26], self-care [29], child and adolescent health [27], fertility [30], cystic fibrosis [31], and gestational diabetes [34,37]. Four studies [6,7,30,35] were methodological studies each applied a structured framework to reviews of specific topics. The remaining described the development of methodological approaches.

Of the 14 frameworks, four [27,30,35,38] were published before the release of the GRADE approach in the first version of the Cochrane Handbook [39] and GRADE BMJ series in 2008 [40]. Among the 10 frameworks developed after the release of GRADE, five [6,7,13,29] explicitly integrated GRADE CoE assessments into their processes. Robinson et al [19] integrated a GRADE-like grading system.

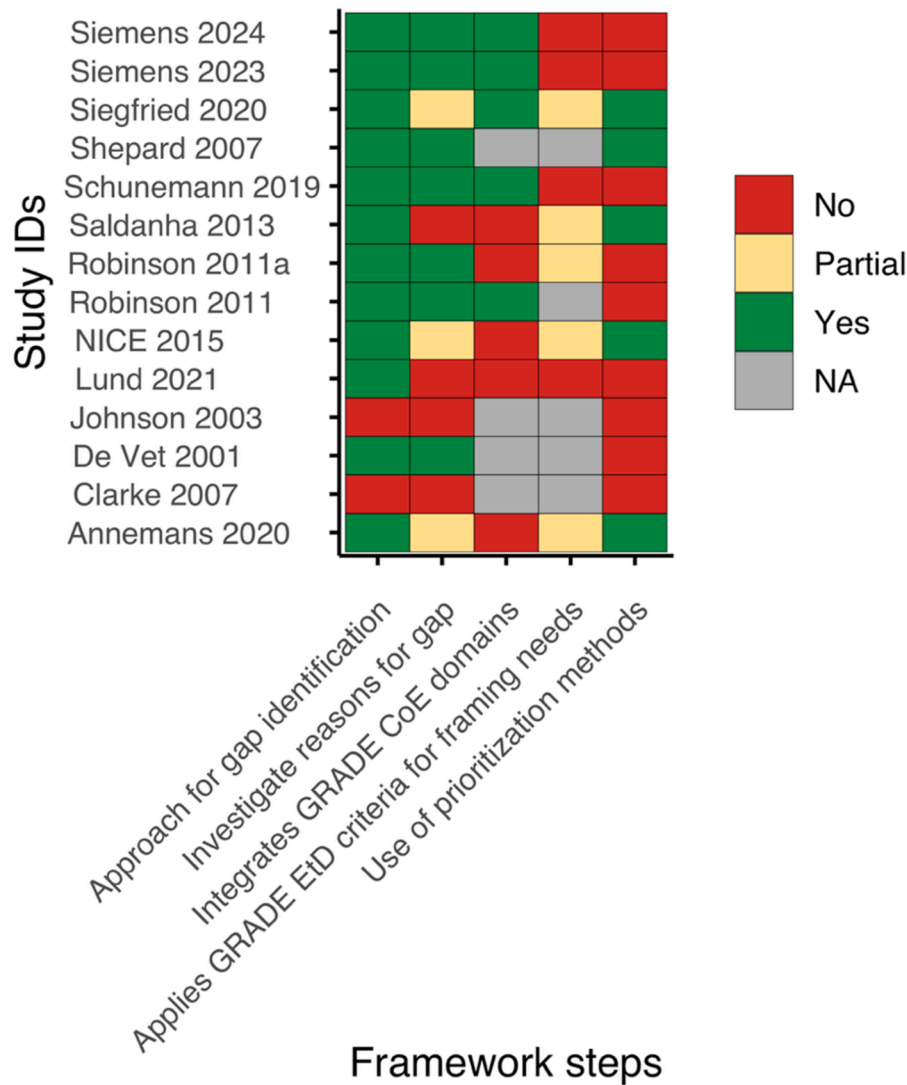
### 3.2. Methods to identify the state of research and research gaps

The included studies employed a variety of structured approaches for identifying research gaps (Fig 3).

Five [6,7,19,31,37] frameworks used Population, Interventions, Comparisons and Outcomes (PICO) to identify the research gaps and provide details about the research without incorporating additional evaluative criteria. Three [26,28,29] proposed identifying research gaps based on uncertainty in the evidence base. One [13] combined PICO to characterize the research gaps with direct linkage to GRADE CoE assessments to conclude about implications for research. Lund et al [36] proposed defining research gaps by the absence of target high-certainty evidence or by situations where conclusions could not be drawn from the available evidence base. One framework [38] proposed identifying research gaps based on the quantity of available trials, while one [27] used a care pathway as the organizing framework. Two approaches [30,35] were unclear on how gaps should be identified.

### 3.3. Methods to classify the nature of the research gaps

Six [6,7,19,27,31,38] frameworks provided explicit methods to guide users in classifying the nature of the a research gap. Three frameworks [26,28,29] did not clearly distinguish between the process of identifying a gap and



**Figure 3.** Heatmap of framework steps for developing research priorities. CoE, certainty of evidence; EtD, evidence to decision; NA, Not applicable; GRADE, Grading of Recommendations Assessment, Development and Evaluation (frameworks published before the introduction of GRADE CoE or the EtD framework). Robinson 2011a and Saldanha 2013 are coded yellow for EtD criteria because, although predating the formal GRADE EtD framework, they considered relevant criteria. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

classifying its nature. In contrast, three frameworks [6,7,19] embedded this process within the GRADE approach, using the CoE domains to determine the cause of the gap. One framework [31] specified two primary natures that could underlie a research gap: insufficient or absent evidence to inform recommendations, and topics needing further research as identified during guideline development. Shepard et al [27] advised reviewing the state of relevant guidelines and evaluation research, as well as the quality and currency of the studies. De Vet et al [38] recommended assessing inconsistency among trial results to determine the nature of the gap.

3.4. Integration of GRADE CoE

Four frameworks [27,30,35,38] predated the first version of the Cochrane Handbook [39] and GRADE BMJ series in 2008 [40]. However, one of these frameworks [38] considered inconsistency, while another [27] considered limitations in studies’ design or execution. Five frameworks [6,7,13,19,29] integrated all five GRADE CoE domains for downgrading the certainty in the evidence—limitations in studies’ design or execution, imprecision, inconsistency, indirectness, and dissemination bias—into their processes. One of those tools was published on behalf of the Cochrane GRADEing methods

**Table 2.** Proposed table for identifying research gaps, framing needs and establishing research priorities

| PICO research question (from SoF) | Nature of the research gap (tick all that apply)                                                                                                                                                                                                                 | Related GRADE CoE domains (tick all that apply)                                                                                                                                                                                                                                                                      | EtD considerations linked to the gap (tick all that apply)                                                                                                                                                                                                                                                                                                                          | Optimal study design needed                                                                                                                           | Priority classification                                                 |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
| PICO 1                            | <input type="checkbox"/> Moderate, low or very low CoE<br><input type="checkbox"/> Outdated evidence<br><input type="checkbox"/> Conditional recommendation based on research gaps <sup>a</sup><br><input type="checkbox"/> Evidence absence → No recommendation | <input type="checkbox"/> Limitations in study design or execution (risk of bias)<br><input type="checkbox"/> Inconsistency<br><input type="checkbox"/> Indirectness (external validity, transferability)<br><input type="checkbox"/> Imprecision<br><input type="checkbox"/> Dissemination (publication) bias<br>N/A | <input type="checkbox"/> Desirable health effects<br><input type="checkbox"/> Undesirable health effects<br><input type="checkbox"/> Values and preferences<br><input type="checkbox"/> Resource use<br><input type="checkbox"/> Equity<br><input type="checkbox"/> Acceptability<br><input type="checkbox"/> Feasibility<br><input type="checkbox"/> Planetary health <sup>b</sup> | <input type="checkbox"/> Randomized or nonrandomized intervention studies <sup>c</sup><br><input type="checkbox"/> Observational studies <sup>d</sup> | <input type="checkbox"/> Yes<br><input type="checkbox"/> No<br>Rank____ |

CoE, certainty of evidence; EtD, evidence to decision; GRADE, Grading of Recommendations Assessment, Development and Evaluation; PICO, Population, Intervention, Comparator, Outcome; RCT, randomized controlled trial.

Please note that the table has yet to undergo rigorous testing and will undergo iterative development based on user experiences and working group discussions.

<sup>a</sup> Conditional recommendations in GRADE indicate that the desirable and undesirable effects are closely balanced or that there is uncertainty in the evidence. When such recommendations arise specifically from research gaps (eg, unresolved subgroup effects), these can sufficiently inform a research recommendation.

<sup>b</sup> GRADE has a guidance on considering planetary health in press the Evidence to Decision Framework in press and published on it and described the methods to its development and application in prior publications [41–44].

<sup>c</sup> For technologies such as pharmaceuticals, diagnostic tests, medical devices, and management strategies.

<sup>d</sup> For other research questions, such as prognostic, etiologic, or contextual factors.

group and includes a simple table describing the state of research across the GRADE CoE domains and describing respective implications [13]. Robinson et al [31] and the NICE approach [28] did not explicitly refer to GRADE but included domains consistent with the GRADE CoE approach in the decision-making process.

### 3.5. Methods to identify and frame research needs

Six frameworks [26,28,29,31,37] incorporated selected criteria that are included in the standard EtD framework into their processes to articulate research needs. Feasibility was assessed in all of them. Desirable and undesirable health effects and values and preferences were considered in five of them [26,29,31,37]. Only TRUST4RD framework [26] incorporated the importance of the problem and acceptability. Equity was addressed in two approaches [28,29].

### 3.6. Integration of GRADE EtD criteria

No framework or tool explicitly integrated the GRADE EtD framework, acknowledging that 57% (8/14) of frameworks were developed prior to publication of the GRADE EtD framework.

### 3.7. Methods to prioritize research needs

Five frameworks described methods for prioritizing research needs [26–29,37]. Approaches included stakeholder engagement [28], Delphi consensus [37], iterative panel discussions [29], workshops [26,27], or methods like deliberative scoring and quantitative modeling [26].

### 3.8. Preliminary tool for identifying research gaps

To facilitate a systematic approach for identifying research gaps, we developed a preliminary tool (Table 2). The structure builds upon the conventional SoF table, with one row corresponding to each original review PICO question. For each entry, authors are prompted to characterize the gap by selecting one or more predefined categories that justify its designation as a research gap (eg, inconsistency, imprecision, indirectness, or lack of evidence). When applicable, the gap is explicitly linked to the relevant GRADE certainty domains.

To determine whether the identified gap warrants further research, authors then map it to the EtD framework criteria, ensuring that the identified need is directly relevant to policymaking and clinical decision-making. The tool subsequently guides users to specify the optimal study design to address the gap and to assess its importance by rating both whether it should be prioritized and, if so, the degree of priority.

This tool is intended as a preliminary framework. It will undergo iterative refinement and empirical testing to evaluate usability, consistency across users, and validity in distinguishing between minor evidence uncertainties and substantive research needs.

## 4. Discussion

This scoping review mapped frameworks for identifying research gaps, translating them into research needs, and establishing priorities from evidence syntheses. Most frameworks outlined multistep approaches that focused on

describing research needs, but few offered guidance on how to determine the underlying rationale for a research gap, specifically, what qualifies as a gap in the first place. Only a few frameworks published after the GRADE approach was published explicitly incorporated GRADE CoE domains, and very few provided methods for prioritizing research once needs were identified.

The review highlighted substantial variability across frameworks. Some presented structured, sequential steps, whereas others were less clear in defining or ordering the process, making operationalization in practice difficult. Importantly, none of the frameworks referenced GRADE as a basis for setting research priorities, despite its widespread use for assessing evidence and informing decisions. These findings align with prior literature, which has shown that research gaps in systematic reviews and guidelines are inconsistently defined, insufficiently described, and often unclearly derived [4–7] (manuscript #2 from project in process). This inconsistency limits the utility of these sections for guiding funding, policy, and research agendas.

#### 4.1. Strengths and limitations

A key strength of this review is the use of rigorous and transparent review methodology, ensuring comprehensive identification of relevant frameworks. The project was conducted within the context of the GRADE Project Group for research, an international subgroup of over 65 experts of the GRADE working group, which provided diverse methodological perspectives and domain expertise. Although we searched MEDLINE, PsycINFO, and Web of Science, we did not search Embase which may be viewed as a limitation. The decision to not include Embase was based on potential feasibility concerns, as its inclusion would result in a substantial increase in the volume of citations. The review was limited by the heterogeneity of the included frameworks, reflecting our decision to adopt a broad interpretation of “framework.” This inclusive approach enhanced comprehensiveness but made direct comparisons and mapping across approaches more challenging. In addition, more than half of the included studies were meta-research projects that examined whether and how systematic reviews addressed implications for research, rather than attempts to develop formal frameworks for setting research priorities. Although not frameworks in the strict sense, we included these studies because their evaluations implicitly assume that research gaps should be systematically reported and addressed, thereby reflecting methodological expectations rather than ad hoc practices. Another limitation was reliance on published descriptions, as some frameworks may have contained details not reported in the included documents. We may also have missed relevant approaches if they were unpublished, embedded within broader guidance, or not explicitly labeled as research gap

identification frameworks. A formal gray literature search was not conducted, as we determined that rigorously developed frameworks were unlikely to exist outside the published peer reviewed literature. However, we did prompt GRADE working group and evidence-based research collaboration members who are experts from diverse global organizations such as WHO and NICE to suggest any relevant resources that were not database indexed. However, none were identified.

#### 4.2. Implications for policy and practice

Our results suggest a clear need for guidance to help authors systematically develop recommendations for further research. Although multiple approaches are available, transparency in the process of identifying gaps, articulating the rationale for prioritization, and reporting research needs remains limited. Greater clarity in these steps would enhance the practical utility of outputs for funders, policy-makers, and researchers. The absence of detailed, standardized procedures has likely posed challenges for implementation. Furthermore, only a small number of existing tools were embedded within internationally adopted frameworks for guideline development, such as the GRADE approach. A structured tool, such as the preliminary table we developed (Table 2), can support authors in applying a systematic approach, linking gaps to GRADE certainty domains, relevance to GRADE EtD criteria, and ultimately ensuring that research priorities are clearly conveyed to relevant users. Without such processes, opportunities to connect evidence syntheses directly to future research agendas may be lost. The findings of this scoping review will also contribute to the development of official guidance by the GRADE working group, led by this project team.

#### 4.3. Implications for research

The variability and lack of prioritization methods highlight a gap in methodological research. Prioritization requires criteria or principles to help guide the process and must be feasible to conduct, less resource-intensive than large-scale priority setting exercises. Future methodological work should focus on developing, testing, and validating a standardized process for identifying, documenting, and prioritizing research gaps from evidence syntheses. As an interim step, we propose a preliminary table of prompts (Table 2), informed by our scoping review, to guide the systematic characterization of gaps and their translation into research needs. This table is provisional and will undergo further development, testing, and formalization through the GRADE development process over the next one to 2 years, before being finalized as official GRADE guidance [25].

## 5. Conclusions

Frameworks for identifying research gaps from evidence syntheses are diverse, poorly and inconsistently structured, and often lack explicit methods for identifying needs and prioritization. Standardized guidance is necessary to enhance clarity, operationalizability, and utility for informed decision-making. The draft tool proposed here represents an initial step toward achieving this goal. This scoping review will inform the development of official guidance by the GRADE working group.

## CRedit authorship contribution statement

**Omar Dewidar:** Writing – original draft, Visualization, Validation, Methodology, Conceptualization. **Mouad Elganga:** Writing – review & editing, Methodology. **Martin Ringsten:** Writing – review & editing, Methodology, Investigation. **Miranda Langendam:** Writing – review & editing, Methodology, Investigation. **K.M. Saif-Ur-Rahman:** Writing – review & editing, Methodology, Investigation. **Malgorzata M. Bala:** Writing – review & editing, Methodology, Investigation. **Chiara Arienti:** Writing – review & editing, Methodology. **Antonio Bognanni:** Writing – review & editing, Methodology. **Andrea J. Darzi:** Writing – review & editing, Methodology. **Paweł Jemioł:** Writing – review & editing, Methodology. **Sebastian Krawczyk:** Writing – review & editing, Methodology. **Stefano G. Lazzarini:** Writing – review & editing, Methodology. **Manuel Marques-Cruz:** Writing – review & editing, Methodology. **Joerg J. Meerpohl:** Writing – review & editing, Methodology. **Getrud J. Mollel:** Writing – review & editing, Methodology. **Elena Parmelli:** Writing – review & editing, Methodology. **Karen A. Robinson:** Writing – review & editing, Methodology. **Bernardo Sousa-Pinto:** Writing – review & editing, Methodology. **Waldemar Siemens:** Writing – review & editing, Methodology. **Rafael José Vieira:** Writing – review & editing, Methodology, Conceptualization. **Samuel J. White:** Writing – review & editing, Methodology. **Maura Marcucci:** Writing – review & editing, Methodology. **Holger J. Schünemann:** Writing – review & editing, Supervision, Project administration, Methodology, Conceptualization.

## Declaration of competing interest

The authors have no declarations to declare.

## Data availability

Data will be made available on request.

## References

- [1] Tenny S, Varacallo MA. Evidence-Based medicine. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing. 2025. Available at: <http://www.ncbi.nlm.nih.gov/books/NBK470182/>. Accessed August 15, 2025.
- [2] Oliver K, Pearce W. Three lessons from evidence-based medicine and policy: increase transparency, balance inputs and understand power. *Palgrave Commun* 2017;3(1):43. <https://doi.org/10.1057/s41599-017-0045-9>.
- [3] NCCIH. Clinical Practice Guidelines. Available at: <https://www.nccih.nih.gov/health/providers/clinicalpractice>. Accessed August 15, 2025.
- [4] Nyanchoka L, Tudur-Smith C, Thu VN, Iversen V, Tricco AC, Porcher R. A scoping review describes methods used to identify, prioritize and display gaps in health research. *J Clin Epidemiol* 2019; 109:99–110. <https://doi.org/10.1016/j.jclinepi.2019.01.005>.
- [5] Wong EC, Maher AR, Motala A, Ross R, Akinniranye O, Larkin J, et al. Methods for identifying health research gaps, needs, and priorities: a scoping review. *J Gen Intern Med* 2022;37(1):198–205. <https://doi.org/10.1007/s11606-021-07064-1>.
- [6] Siemens W, Bantle G, Ebner C, Blümle A, Becker G, Schwarzer G, et al. Evaluation of “implications for research” statements in systematic reviews of interventions in advanced cancer patients - a meta-research study. *BMC Med Res Methodol* 2023;23(1):302. <https://doi.org/10.1186/s12874-023-02124-y>.
- [7] Siemens W, Bantle G, Mahler S, Nothacker J, Stadelmaier J, Bitzer EM, et al. Clinical and methodological implications for research elements in systematic reviews on COVID-19 treatment were often unstructured and under-reported: a meta-research study. *J Clin Epidemiol* 2024;166: 111236. <https://doi.org/10.1016/j.jclinepi.2023.111236>.
- [8] Priority Setting Partnership | NIHR JLA. Available at: <https://www.jla.nihr.ac.uk/priority-setting-partnerships>. Accessed August 15, 2025.
- [9] Rudan I, Gibson JL, Ameratunga S, El Arifeen S, Bhutta ZA, Black M, et al. Setting priorities in global child health research investments: guidelines for implementation of CHNRI method. *Croat Med J* 2008;49(6):720–33. <https://doi.org/10.3325/cmj.2008.49.720>.
- [10] Guyatt G, Oxman AD, Akl EA, Kunz R, Vist G, Brozek J, et al. GRADE guidelines: 1. Introduction-GRADE evidence profiles and summary of findings tables. *J Clin Epidemiol* 2011;64(4):383–94. <https://doi.org/10.1016/j.jclinepi.2010.04.026>.
- [11] Alonso-Coello PAD, Moberg J, Petersen B, Romina A, Elie A, Davoli M, et al. GRADE Evidence to Decision (EtD) frameworks: a systematic and transparent approach to making well informed health-care choices. 2: clinical practice guidelines. *BMJ* 2016;353(NA): i2089–NA. <https://doi.org/10.1136/bmj.i2089>.
- [12] GRADE book. Available at: <https://book.grade.pro.org/>. Accessed September 5, 2025.
- [13] Schünemann HJ, Elisabeth VG, Higgins JPT, Santesso N, Deeks JJ, Glasziou P, et al. Cochrane Handbook for Systematic Reviews of Interventions - interpreting results and drawing conclusions. *Cochrane Handbook Syst Rev Interv* 2019;403. <https://doi.org/10.1002/9781119536604.ch15>.
- [14] Dewidar O, Schünemann H. Research recommendations in reviews and guidelines 2025. <https://doi.org/10.17605/OSF.IO/DFWBZ>.
- [15] Peters MD, Godfrey C, McInerney P, Munn Z, Tricco AC, Khalil H. Scoping reviews. In: Aromataris E, Lockwood C, Porritt K, Pilla B, Jordan Z, editors. *JBIM Manual for Evidence Synthesis*. JBI. Available at: <https://jbi-global-wiki.refined.site/space/MANUAL/355862497/10.+Scoping+reviews>. Accessed April 28, 2025.
- [16] Tricco AC, Lillie E, Zarin W, O'Brien KK, Colquhoun H, Levac D, et al. PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Ann Intern Med* 2018;169(7):467–73. <https://doi.org/10.7326/M18-0850>.
- [17] Zhao J, Chen W, Bai W, Zhang X, Hui R, Chen S, et al. Research priority setting for implementation science and practice: a living systematic review protocol. *Syst Rev* 2025;14(1):51. <https://doi.org/10.1186/s13643-025-02786-3>.
- [18] Robinson KA, Saldanha JJ, McKoy NA. Frameworks for Determining Research Gaps During Systematic Reviews. *Frameworks*

- for Determining Research Gaps During Systematic Reviews Rockville, MD: Agency for Healthcare Research and Quality. 2011. Available at: [https://effectivehealthcare.ahrq.gov/sites/default/files/pdf/methods-future-research-steps-framework\\_research.pdf](https://effectivehealthcare.ahrq.gov/sites/default/files/pdf/methods-future-research-steps-framework_research.pdf). Accessed May 13, 2026.
- [19] Robinson KA, Saldanha IJ, McKoy NA. Development of a framework to identify research gaps from systematic reviews. *J Clin Epidemiol* 2011;64(12):1325–30. <https://doi.org/10.1016/j.jclinepi.2011.06.009>.
  - [20] Haddaway NR, Grainger MJ, Gray CT. citationchaser: an R package for forward and backward citations chasing in academic searching. Available at: <https://github.com/nealhaddaway/citationchaser>. Accessed August 19, 2025.
  - [21] Bramer WM, Giustini D, de Jonge GB, Holland L, Bekhuis T. Deduplication of database search results for systematic reviews in EndNote. *J Med Libr Assoc* 2016;104(3):240–3. <https://doi.org/10.3163/1536-5050.104.3.014>.
  - [22] Scott NA, Moga C, Harstall C, Magnan J. Using health technology assessment to identify research gaps: an unexploited resource for increasing the value of clinical research. *Health Policy* 2008;3(3):e109–27.
  - [23] Patel HD, Iyoha E, Pierorazio PM, Sozio SM, Johnson MH, Sharma R, et al. A systematic review of research gaps in the evaluation and management of localized renal masses. *Urology* 2016;98:14–20.
  - [24] Wickham H. *ggplot2: Elegant Graphics for Data Analysis*. New York: Springer-Verlag; 2016.
  - [25] Schünemann HJ, Brennan S, Akl EA, Hultcrantz M, Alonso-Coello P, Xia J, et al. The development methods of official GRADE articles and requirements for claiming the use of GRADE – a statement by the GRADE guidance group. *J Clin Epidemiol* 2023;159:79–84. <https://doi.org/10.1016/j.jclinepi.2023.05.010>.
  - [26] Annemans L, Makady A. TRUST4RD: tool for reducing uncertainties in the evidence generation for specialised treatments for rare diseases. *Orphanet J rare Dis* 2020;15(1):127. <https://doi.org/10.1186/s13023-020-01370-3>.
  - [27] Shepherd J, Briggs J, Payne L, Packer C, Kerridge L, Ashton-Key M. Setting the future policy agenda for health technology assessment: a specialty mapping approach. *Int J Technol Assess Health Care* 2007;23(4):405–13. <https://doi.org/10.1017/S0266462307070638>.
  - [28] National Institute for Health and Care Excellence. Research Recommendations Process and Methods Guide. 2015. Available at: [ResearchRecommendationsProcessandMethodsGuide](https://www.nice.org.uk/research-recommendations-process-and-methods-guide). Accessed August 15, 2025.
  - [29] Siegfried N, Narasimhan M, Logie CH, Thomas R, Ferguson L, Moody K, et al. Prioritising gender, equity, and human rights in a GRADE-based framework to inform future research on self care for sexual and reproductive health and rights. *BMJ Glob Health* 2020;5(3):e002128. <https://doi.org/10.1136/bmjgh-2019-002128>.
  - [30] Johnson NP, Proctor M, Farquhar CM. Gaps in the evidence for fertility treatment-an analysis of the Cochrane Menstrual Disorders and Subfertility Group database. *Hum Reprod* 2003;18(5):947–54.
  - [31] Robinson KA, Saldanha IJ, McKoy NA. Identification of research gaps from evidence-based guidelines: a pilot study in cystic fibrosis. *Int J Technol Assess Health Care* 2011;27(3):247–52. <https://doi.org/10.1017/S0266462311000225>.
  - [32] Robinson KA, Akinyede O, Dutta T, Sawin VI, Li T, Spencer MR, et al. Framework for Determining Research Gaps During Systematic Review: Evaluation. Rockville (MD): Agency for Healthcare Research and Quality (US). 2013. Available at: <http://www.ncbi.nlm.nih.gov/books/NBK126708/>. Accessed August 15, 2025.
  - [33] Bennett WL, Nicholson WK, Saldanha IJ, Wilson LM, McKoy NA, Robinson KA. Results. In: Future Research Needs for the Management of Gestational Diabetes Agency for Healthcare Research and Quality (US). 2010. Available at: <https://www.ncbi.nlm.nih.gov/books/NBK51288/>. Accessed August 15, 2025.
  - [34] Bennett WL, Robinson KA, Saldanha IJ, Wilson LM, Nicholson WK. High priority research needs for gestational diabetes mellitus. *J Women's Health* (2002) 2012;21(9):925–32. <https://doi.org/10.1089/jwh.2011.3270>.
  - [35] Clarke L, Clarke M, Clarke T. How useful are Cochrane reviews in identifying research needs? *J Health Serv Res Policy* 2007;12(2):101–3. <https://doi.org/10.1258/135581907780279648>.
  - [36] Lund H, Juhl CB, Nørgaard B, Draborg E, Henriksen M, Andreasen J, et al. Evidence-Based Research Series-Paper 3: using an Evidence-Based Research approach to place your results into context after the study is performed to ensure usefulness of the conclusion. *J Clin Epidemiol* 2021;129:167–71. <https://doi.org/10.1016/j.jclinepi.2020.07.021>.
  - [37] Saldanha IJ, Wilson LM, Bennett WL, Nicholson WK, Robinson KA. Development and pilot test of a process to identify research needs from a systematic review. *J Clin Epidemiol* 2013;66(5):538–45. <https://doi.org/10.1016/j.jclinepi.2012.07.009>.
  - [38] de Vet HC, Kroese ME, Scholten RJ, Bouter LM. A method for research programming in the field of evidence-based medicine. *Int J Technol Assess Health Care* 2001;17(3):433–41. <https://doi.org/10.1017/s0266462301106148>.
  - [39] Schünemann HJ, Oxman AD, Vist GE, Higgins JP, Deeks JJ, Glasziou P, et al. Interpreting results and drawing conclusions. In: *Cochrane Handbook for Systematic Reviews of Interventions*: 359–87. <https://doi.org/10.1002/9780470712184.ch12>.
  - [40] Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ* 2008;336(7650):924–6. <https://doi.org/10.1136/bmj.39489.470347>.
  - [41] Vieira RJ, Sousa-Pinto B, Herrmann A, Antó J, Williams S, et al. A novel approach to consider Planetary Health in Guideline development: a GRADE approach using the allergic rhinitis and its impact on asthma (ARIA) 2024-2025 Guidelines as a case Study. *J Allergy Clin Immunol In Pract* 2025;13:2600–7.
  - [42] Piggott T, Raja M, Michels CTJ, Herrmann A, Scallan KA, Darzi AJ, et al. Considering planetary health in health guidelines and health technology assessments: a scoping review protocol. *Syst Rev* 2024;13:163.
  - [43] Saadat P, Herrmann A, Sousa-Pinto B, Michels CTJ, Thomson D, Senerth E, et al. Planetary health in health guidelines: expert workshop report 2024. *Clin Public Health Guidel* 2026;3:e70072.
  - [44] Piggott T, Saadat P, Herrmann A, Tetelbom Stein A, Darzi AJ, Haines A, et al. Integrating Planetary Health in Health Guidelines (GRADE Guidance 46). *Ann Intern Med* 2026;12. <https://doi.org/10.7326/ANNALS-25-04761>.